

Request for Information: Enterprise Dental Imaging PACS Solution (Enhanced Version)

Attachment 1: Defense Health Agency (DHA) Enterprise Dental Picture Archiving and Communication System (PACS)

Background: The Defense Health Agency (DHA) seeks information from industry regarding solutions, approaches, and best practices for deploying a secure, scalable, and vendor-neutral enterprise Picture Archiving and Communication System (PACS) for dental imaging. The goal is to modernize and standardize the capture, storage, retrieval, display, and sharing of all dental imaging data across the global DHA network, ensuring seamless integration with the Military Health System (MHS) GENESIS Electronic Health Record (EHR) and support a full spectrum of dental services, including veterinary care.

Challenges: The deployment of a true enterprise dental PACS capability presents significant and complex challenges. A successful solution must overcome the following:

- **Interoperability in a Heterogeneous Environment:** Operating with a wide array of dental imaging modalities and hardware from various vendors, while ensuring compliance with DICOM and IHE standards and seamless integration with the MHS GENESIS EHR and Enterprise Clinical Imaging Archive (ECIA).
- **Cybersecurity and Access Control:** Adhering to the Department of War's (DoW) rigorous cybersecurity requirements, including Zero-Trust Architecture (ZTA), NIST guidelines, and DISA STIGs, while implementing robust, user-friendly access controls (CAC/PIV/SSO) that do not impede critical clinical workflows.
- **Performance in Disconnected Environments:** Supporting a high volume of daily patient encounters (up to 7,500 enterprise-wide) and massive data objects (up to 2GB per patient) across a global network with varying bandwidth, including fully disconnected, low-connectivity, and no-comm environments, while ensuring rapid image loading times and reliable background data synchronization.
- **Data Integrity and Lifecycle Management:** Ensuring 100% accurate patient data matching and synchronization from authoritative sources, managing data migration from legacy systems (e.g., MiPACS, Apteryx, VistaA), enforcing customizable business rules to protect data integrity at the point of capture, and handling complex patient name variances.
- **Clinical and Specialized Workflow Adaptation:** Supporting multiple complex clinical workflows, including order-based and encounter-based imaging, while providing customizable templates, advanced image manipulation tools, chairside CAD/CAM integration, and specific workflows for specialties like periodontics, endodontics, and even veterinary dentistry (canine, feline, equine).
- **Centralized Management and Support:** Managing a globally distributed system from a central point, providing timely technical support, delivering effective "train-the-trainer"

programs, and providing robust administrative reporting on system usage and data quality.

Objectives: The DHA seeks to achieve the following high-level objectives through this enterprise dental imaging solution:

- **Enhance Clinical Efficiency and Diagnostic Quality:** Provide a streamlined, reliable, and vendor-neutral workflow that accelerates image acquisition and access, while offering high-fidelity viewing, manipulation, and comparison tools to improve diagnostic accuracy and support advanced treatment planning.
- **Ensure a Unified and Interoperable Health Record:** Enable flawless integration with MHS GENESIS, ECIA, and all standard dental modalities, ensuring imaging data is a readily accessible component of the patient's longitudinal health record.
- **Strengthen Cybersecurity and Data Protection:** Implement a solution that meets or exceeds all DoW cybersecurity and access control standards, guaranteeing the protection of patient data and system integrity through auditable, centrally managed controls.
- **Achieve Enterprise Scalability and Governance:** Deploy a solution that can scale across all CONUS and OCONUS sites, including those with limited infrastructure, while enabling centralized system management, configuration, reporting, and enforcement of clinical standards. **There are approximately 170 Medical Treatment Facilities (MTF)s and 600+ Dental Treatment Facilities (DTFs).**
- **Support Future Innovation and Advanced Analytics:** Establish a flexible platform capable of supporting future Artificial Intelligence (AI) and Machine Learning (ML) applications for clinical decision support, population health initiatives, and automated quality reviews.

Specific Tasks

1.0 System Architecture & Scalability

1.1 Enterprise Load: System must support up to 7,500 patients per day across the enterprise, with data volumes from 1 Kilobyte to 2GB per patient. **Anticipated migration includes roughly 1GB to 8TB or more of data per site, depending on the size and scope of the facility.**

1.2 Disconnected Operations: System must feature a local database that can be fully accessed without server-class software (e.g., SQL Server) for use in no-comm areas. It must be capable of storing a minimum of 3 years of data (for ~10,000 patients) and automatically sync data in the background upon network connection.

1.3 Virtualization: System must be capable of operating in a virtual environment like Docker, VMWare, etc.

1.4 Architecture: The architecture must support a hub-and-spoke model to minimize hardware/software requirements at smaller satellite facilities.

1.5 Cloud Capability: The solution should be cloud-enabled, which must support a hybrid model that stores images both locally and in a cloud environment, accessible via a dental-specific FDA-approved web viewer.

2.0 EHR & System Integration

2.1 EHR Integration: System must interact with the DoD and VA EHR (MHS GENESIS) to acquire and synchronize patient demographic data and must be launchable directly from the EHR, passing patient context automatically.

2.2 DICOM & IHE: System must be DICOM 3.0 compliant, support JPEG2K compression, and adhere to IHE protocols. This includes performing DICOM queries (C-FIND, C-GET, C-MOVE) to the Enterprise Clinical Imaging Archive (ECIA). **Solutions must store all medical images in non-proprietary DICOM format.**

2.3 ECIA Archiving: System must use the ECIA for long-term archiving and ensure its native data structure is fully compatible.

2.4 Workflow Management: System must utilize unique accession numbers for both order-based and encounter-based imaging workflows.

2.5 Data Migration: System must provide a clear methodology for migrating data from legacy systems, including VistA and physical media backups (e.g., CDs).

2.6 API Support: System must support third-party integration via APIs and/or C-commands for direct communication and manipulation.

3.0 Modality & Vendor Integration

3.1 Vendor Neutrality: System must be vendor-neutral, integrating with multiple dental hardware vendors using TWAIN, SDKs, or drivers.

3.2 Modality Support: System must capture, store, and view images from all standard dental modalities, including: Intra-Oral (IO), Panoramic (PX), CBCT, visible light (Intra-Oral Cameras, face scans), and Computed Radiography (CR).

3.3 Raw Data Interaction: System must be capable of interacting with the least-processed data from modalities, including reading proprietary data structures.

3.4 Non-Proprietary Storage: System must capture and store images in non-proprietary formats (e.g., .DIC, DCM).

3.5 Third-Party Viewers: System must have the ability to bridge to and launch specific third-party CBCT viewers based on the image manufacturer (e.g., Planmeca images open in Romexis).

4.0 Clinical Workflow & User Interface

4.1 Anatomical Display: System must display images anatomically correct using DICOM structured display and allow querying by ADA Universal Tooth Numbering System.

4.2 Customization: System must support customizable progression templates, full-mouth series, and report layouts. The user interface must be scalable to hide or show features based on user role or facility need.

4.3 Image Manipulation: System must provide high-quality, non-destructive image enhancement and manipulation tools (Brightness, Contrast, Zoom, etc.) and allow for side-by-side comparison of historical and current images.

4.4 Treatment Planning: System must include tools for implant treatment planning, allowing providers to overlay scaled implants from various manufacturers on an image. It should also support endodontic workflows by allowing canal tracing.

4.5 Fabrication & Lab Integration: System must facilitate the transmission of images and prosthetic data to dental labs for fabricating appliances (e.g., surgical guides). It must also support chairside storing of CAD/CAM scans.

4.6 Veterinary Support: System must support veterinary dentistry, with the ability to switch tooth charts to display for canine, feline, and equine patients.

4.7 Dictation: System must facilitate integration with third-party dictation software (e.g., Dragon) at the workstation.

4.8 Ai: System must facilitate integration with third-party Ai software that is hosted on prem only. Ai must be FDA approved for at least 18 years of age and up. Ai should be able to detect carries, bone loss, and more.

5.0 Cybersecurity & Data Integrity

5.1 Compliance: System and vendor must comply with NIST 800-53, 800-171, and FIPS 200 guidelines.

5.2 Access Control: System must support Single Sign-On (SSO) using DoD Smart Card/Active Directory authentication.

5.3 Data Validation: System must allow administrators to apply custom business rules to data

entry fields (e.g., Patient ID format) to prevent errors. It must also have robust functionality for handling patient name variances.

5.4 Audit & Reporting: System must provide robust administrative reporting to track image retakes (including user and customizable reasons), data corrections, and other quality metrics. All actions must be logged in a configurable audit trail.

6.0 Support & Training

6.1 Technical Support: Vendor must provide comprehensive technical support for the central system and all connected devices, with clearly defined resolution times.

6.2 Upgrades: Vendor must provide routine bug fixes and version upgrades to enhance capabilities.

6.3 Training: Vendor must conduct user training at each installation and should offer a ‘train-the-trainer’ program to allow for certified, self-sustaining local training.